

JEE MAIN REVISION MASTER: ELECTROMAGNETIC WAVES

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 15 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Displacement Current, EM Spectrum & Wave Properties	15 PYQs	EM Waves

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Displacement Current, EM Spectrum & Wave Properties	$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{E_0}{B_0}$, $I_d = \epsilon_0 \frac{d\Phi_E}{dt}$, $\text{Energy Density: } u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} B^2 \mu_0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Displacement Current, EM Spectrum & Wave Properties (Very High Yield — 15 PYQs)

Core Principle: Displacement current $I_d = \epsilon_0 (d\Phi_E / dt)$. EM waves are transverse with mutually perpendicular $\vec{E}$ and $\vec{B}$ vectors: $\vec{E} \perp \vec{B} \perp \vec{c}$. Speed of propagation $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = (E_0 / B_0) \approx 3 \times 10^8 \text{ m/s}$. EM spectrum in order of increasing wavelength: $\gamma\text{-rays} < \text{X-rays} < \text{UV} < \text{Visible} < \text{IR} < \text{Microwaves} < \text{Radio}$.

■ **Caution / JEE Pitfall:** In an EM wave, the average electric energy density equals the average magnetic energy density: $u_E = u_B$!

■ **Speed Trick:** Mnemonic for decreasing frequency: 'Grandma's eX-box Uses Video In Most Rooms' (Gamma, X-ray, UV, Visible, IR, Micro, Radio).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "What range of electromagnetic spectrum is considered as light?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Match the following Maxwell's equations: (The symbols used here have their usual meanings) List-I List-II (a) Gauss' law for electrostatics ..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Match the following types of waves with their wavelength ranges" $\begin{aligned} &\text{Waves} \\ &\text{Wavelength ranges} \end{aligned}$ [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

What range of electromagnetic spectrum is considered as light?

- (A) 1 mm to 700 nm
- (B) 400 nm to 1 nm
- (C) 400 nm to 700 nm
- (D) 1 nm to 10^{-3} nm

Correct Answer: None

Solution Analysis: The range of the electromagnetic spectrum considered as visible light is approximately 400nm to 700nm. This is the part of the electromagnetic spectrum that the human eye can detect. So, the correct option is: Option C: 400nm to 700nm